

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12386750
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Gruddanti; Srikanth Reddy et al.

Last level cache hierarchy in chiplet based processors

Abstract

An accelerated processor includes a processor core die including a plurality of compute units, the plurality of compute units including a first level (L1) cache. The accelerated processor also includes a plurality of memory cache dies coupled to the processor core die, the plurality of memory cache dies including a last level cache (LLC) such as a level 3 (L3) cache. The accelerated processor includes an LLC controller to issue a cache access request to the LLC and, based on a latency of the cache access request, direct the cache access request to a subset of the plurality of memory cache dies.

Inventors: Gruddanti; Srikanth Reddy (Bangalore, IN), Gummidipudi; Krishnaiah (Bangalore, IN), Vallur; Prasant Kumar (Hyderabad, IN), McIntyre; David Hugh (Santa Clara, CA), Mangaser; Ramon Apostol (Boxborough, MA)

Applicant: ADVANCED MICRO DEVICES, INC. (Santa Clara, CA)

Family ID: 95156481

Assignee: Advanced Micro Devices, Inc. (Santa Clara, CA)

Appl. No.: 18/374757

Filed: September 29, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250110875 A1	Apr. 03, 2025

Publication Classification

Int. Cl.: G06F12/0811 (20160101)

U.S. Cl.:

Field of Classification Search

CPC: G06F (12/0811); G06F (2212/283)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2008/0282032	12/2007	Shen	711/117	G06F 12/0811
2021/0097013	12/2020	Saleh	N/A	G06F 12/0837
2021/0098419	12/2020	Saleh	N/A	H01L 21/563
2024/0272908	12/2023	Liljedahl	N/A	G06F 9/30043
2024/0303195	12/2023	Wu	N/A	G06F 12/0891

Primary Examiner: Rossiter; Sean D

Background/Summary

BACKGROUND

(1) Some processing units include multiple processor cores on a single semiconductor die. The processor cores concurrently execute numerous threads or other workloads to increase the processing unit's throughput. In order to accelerate access to the data required to execute the threads or workgroups, these processing units often utilize a cache hierarchy. The cache hierarchy includes, for example, a local first level (L1) cache at the processor core level, an intermediate second level (L2) cache, and a last level cache (LLC) (e.g., a third level (L3) cache) shared among the processor cores of the processing unit. The cache levels have limited capacities and are often subject to concurrent accesses from a large number of threads or workgroups executing at the processor cores. As such, efficient management and allocation of data across the different levels in the cache hierarchy is one factor in improving processing unit performance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure may be better understood, and its numerous features and advantages made apparent to those skilled in the art by referencing the accompanying drawings. The use of the same reference symbols in different drawings indicates similar or identical items.
- (2) FIG. 1 is an example of a processing system with an accelerated processor configured to employ a LLC hierarchy at the L3 cache, in accordance with various implementations.
- (3) FIGS. 2 and 3 shows examples of an accelerated processor, such as one corresponding to the accelerated processor of FIG. 1, to employ an LLC hierarchy across multiple memory cache dies (MCDs), in accordance with various implementations.
- (4) FIG. 4 shows an example configuration for populating memory cache die positions in an application processor with a reduced LLC size, in accordance with various implementations.
- (5) FIG. 5 is a flow diagram illustrating a method for creating and utilizing non-uniform cache

access (NUCA) techniques at an accelerated processor, in accordance with various implementations.

DETAILED DESCRIPTION

(6) Processing units such as accelerated processors typically include a large number of processor cores (also referred to as compute units (CUs)) that are configured to execute operations concurrently or in parallel. The operations are, in some cases, associated with an application running on a central processing unit (CPU) or other host processor in a processing system including the accelerated processor. In various scenarios, the accelerated processor relies on different bandwidth management techniques to achieve higher throughput for executing the operations at its processor cores. These bandwidth management techniques include managing the allocation of and access to data across the different levels (e.g., the L1, L2, and L3 caches) of the cache hierarchy. In general, when executing operations at multiple ones of its processor cores, the accelerated processor experiences higher latencies for LLC accesses since the LLC cache (e.g., an L3 cache) is shared among multiple processor cores. Furthermore, in some processing units, the LLC is located on multiple memory cache dies (also referred to herein as chiplets) that are separate from the processor core die on which the processor cores and higher levels of cache (e.g., the L1 cache and L2 cache) are located. These memory cache dies (MCDs) typically have different die-to-die interface lengths with the processor core die. Conventional cache management approaches for these types of configurations do not account for the different die-to-die interface lengths between the processor core die and the multiple MCDs. Accordingly, the higher LLC latencies become more noticeable in accelerated processor configurations implementing the LLC on MCDs due to the different die-to-die interface lengths between the MCDs and the processor core die. FIGS. 1-5 illustrate techniques that leverage the different die-to-die interface lengths between the LLC MCDs and the processor core die to implement a non-uniform cache access (NUCA) technique at the LLC to improve LLC latency times, thereby improving system performance.

(7) To illustrate, in some implementations an accelerated processor includes a processor core die with a plurality of CUs. The processor core die includes a first level cache (L1 cache) at each of the CUs, and a second level cache (L2 cache) on the processor core die. The accelerated processor further includes multiple MCDs. The MCDs implement a last layer cache (LLC) such as a third level cache (L3 cache). The LLC MCDs are connected to the processor core die through a series of die-to-die interfaces that are routed to the processor core die through a shared physical (PHY) interface point (IP). In some cases, the shared PHY IP includes serializer/deserializer (SerDes) circuitry with a clock domain crossing structure which uses a first in, first out (FIFO) queue for writes and/or reads to the LLC MCDs. The LLC MCDs are grouped or divided into a plurality of subsets of LLC MCDs based on their respective die-to-die interface lengths with the shared PHY IP, and a cache controller at the accelerated processor assigns a different FIFO write-read pointer separation (also referred to as FIFO pointer separation or pointer separation) to each subset. As used herein, the term “subset” is intended to mean a set each of whose elements is an element of an inclusive set. That is, a set A is a subset of set B if all of the elements of set A (which, in some implementations, include only one element) are also elements of set B. For example, a subset of a set of six memory cache dies in some implementations includes one memory cache die of the six memory cache dies, in other implementations includes two memory cache dies of the six memory cache dies, and so on.

(8) For example, if multiple pairs of the LLC MCDs are symmetrically arranged around the processor core die, a first subset of the LLC MCDs corresponds to a first pair of LLC MCDs with the shortest die-to-die interface length, a second subset of the LLC MCDs corresponds to a second pair of LLC MCDs with the next shortest die-to-die interface length, and so on. Based on these groupings and their assigned FIFO pointer separations, the cache controller implements an LLC hierarchy. That is, the first level of the LLC hierarchy is allocated to the first pair of LLC MCDs with the shortest die-to-die interface length (and thus, the lowest FIFO pointer separation), a second

level of the LLC hierarchy is allocated to the second pair of LLC MCDs with the next shortest die-to-die interface length, and so on. The cache controller allocates critical applications or data to the first level of the LLC hierarchy and allocates non-critical applications or data to the second or lower levels of the LLC hierarchy. This reduces LLC latencies for critical applications or data since these types of applications or data are allocated to LLC MCDs with shorter round trip times between the processor core die and the respective LLC MCD. This, in turn, reduces the overall cache latency in the accelerated processor, thereby resulting in improved processing performance.

(9) In some implementations, for a given supply voltage and frequency of operation, the techniques described herein translate to approximately 4 to 5 core clock cycles of latency improvement at the L3 cache. Considering that a typical average cache latency amounts to 60 core clock cycles, the average latency improvement is improved by about 6-8%. This results in an instructions per cycle (IPC) improvement of about 1% or more for all memory latency bound applications, which is significant for real time workloads or high performance computing (HPC) applications.

(10) In further implementations, the techniques described herein are used in die harvesting to identify an order or preference for which MCD (i.e., L3 cache chiplet) positions are populated in processing units with a reduced L3 cache. For example, by utilizing the techniques described herein, the MCDs at positions associated with shorter latency times are implemented in processing units having fewer MCDs while the other MCD positions are left empty. This improves the performance of processing units with a reduced L3 cache size.

(11) FIG. 1 illustrates an example of a processing system **100** configured to employ a NUCA hierarchy at the LLC in accordance with various implementations. In the illustrated example, the processing system **100** includes a central processing unit (CPU) **102**, an accelerated processor (AP) **104**, a device memory **106** utilized by the AP **104**, and a system memory **108** shared by the CPU **102** and the AP **104**. The memories **106**, **108** include any of a variety of random access memories (RAMs) or combinations thereof, such as a double-data-rate dynamic random access memory (DDR DRAM), a graphics DDR DRAM (GDDR DRAM), and the like.

(12) In some implementations, the AP **104** is a parallel processor that is able to execute a single instruction on a multiple data or threads in a parallel manner. Examples of parallel processors (also referred to as co-processors, vector processors, or the like) include processors such as graphics processing units (GPUs), massively parallel processors, single instruction multiple data (SIMD) architecture processors, machine learning (ML) application-specific integrated circuits (ASICs), and single instruction multiple thread (SIMT) architecture processors for performing graphics, machine intelligence or compute operations. In some implementations, the one or more parallel processors are separate devices that are included as part of a computer. In other implementations such as advance processor units, the one or more parallel processors are included in a single device along with a host processor such as a central processor unit (CPU). Although the description herein uses an accelerated processor (AP) for illustration purposes, the implementations and implementations described below are applicable to other types of parallel processors.

(13) As illustrated, the AP **104** includes a memory controller (MC) **110** for managing address translation operations for one or both of the memories **106**, **108** and a plurality of compute units (CUs) **112** (numbered **112(1)**, **112(2)**, **112(3)**, and through **112(N)**). CUs **112** are also interchangeably referred to as AP cores or processor cores herein. In various implementations, the CUs **112** each include one or more SIMD units (not shown) that execute a thread concurrently with execution of other threads in a wavefront, such as according to a single-instruction, multiple-data (SIMD) execution model. In some implementations, the plurality of CUs **112(1)** to **112(N)** is positioned on a processor core die **120**. While one processor core die **120** is shown in FIG. 1 for clarity purposes, in some implementations, the AP **104** includes additional processor core dies **120**.

(14) The MC **110** operates as the interface between the cache hierarchy in the AP **104** and the system memory **108**. Data to be cached in the cache hierarchy typically is manipulated as blocks of data referred to as “cache lines,” and which are addressed or otherwise located in a memory

hierarchy using a physical address of system memory **108**. Cache lines are accessed from the system memory **108** by the memory controller (MC) **110** in response to memory requests from the AP **104**.

(15) The AP **104** further includes a cache hierarchy including one or more levels of data and instruction caching, such as a first level (L1), a second level (L2), a third level (L3) of caches, and the like. Each cache level includes one or more caches at that level. To illustrate, each CU **112** implements a first level of cache such as private level 1 (L1) cache **114**, which are depicted as L1 caches **114(1)**, **114(2)**, **114(3)**, and through **114(N)** (each L1 cache **114** associated with a corresponding one of the CUs **112(1)**, **112(2)**, **112(3)**, and through **112(N)**). Each of the L1 caches **114** is a private cache for the CU **112** that stores a small amount of recently used or frequently accessed data for its associated CU **112**.

(16) The AP **104** also implements a second level of cache such as a level 2 (L2) cache **116** and a third level of cache such as a shared level 3 (L3) cache **118**. The L2 cache **116** is depicted as L2 caches **116(1)** through **116(J)** and **116(K)**. The shared L3 cache **118** is depicted as L3 caches **118(1)**, **118(2)**, and through **118(M)**. The L3 caches **118** are shared by the CUs **112** of the AP **104**, and thus are also shared by at least the L1 caches **114** and L2 caches **116**. In some implementations, the shared L3 cache level represents a last level cache (LLC). The LLC represents the last cache in the cache hierarchy of the AP **104**. Although the illustrated example includes three levels, in other implementations the cache hierarchy includes fewer than three levels or more than three levels of cache. The various levels of the cache hierarchy cache data for access and manipulation by the AP **104**. Typically, caches at a lower level (e.g., L1) tend to have lower storage capacity and lower access latencies, while caches at the higher level (e.g., L3) tend to have higher storage capacity and higher access latencies. Accordingly, cache lines of data are transferred among the caches of different cache levels so as to optimize utilization of the cache data in view of the respective caches' storage capacities and access latencies.

(17) As illustrated, in some implementations, the L3 caches **118** are located separate from the processor core die **120** housing the plurality of CUs **112**, including the L1 caches **114**, and the L2 caches **116**. For example, in some implementations, each of the L3 caches **118** are located on one or more memory cache dies (MCDs) or chiplets that are adjacent to the processor core die **120** in the AP **104**. That is, in some implementations, L3 cache **118(1)** is positioned on a first MCD or chiplet, L3 cache **118(2)** is positioned on a second MCD or chiplet, and so forth.

(18) In some implementations, the plurality of MCDs implementing the L3 caches share a shared PHY interface point **130** with the processor core die **120**. For example, in some configurations, the shared PHY interface point **130** includes a Serializer/Deserializer (SerDes) circuitry with a clock domain crossing structure (not picture in FIG. 1). The clock domain crossing structure, in some implementations, uses a first in, first out (FIFO) queue for writes and/or reads to the MCDs implementing the L3 caches **118**. Due to the many-to-few communication pattern between the CUs **112** and the L3 caches **118** in the AP **104** and the L3 caches **118** being implemented on an MCD (or chiplet) separate from the processor core die **120**, the latencies associated with L3 cache accesses are relatively long compared with the cache access latencies to the L1 caches **114** and L2 caches **116**. Accordingly, as described herein, the processing system **100** leverages the different die-to-die distances between the processor core die **120** and the MCDs with the L3 caches **118** to implement a NUCA hierarchy across the L3 caches **118**. For example, if M=3 in FIG. 1 (i.e., there are three L3 caches **118(1)** to **118(3)**), L3 cache **118(2)** is the closest L3 cache to the shared PHY interface point **130**. Thus, the roundtrip latency for accessing L3 cache **118(2)** is the lowest among the L3 caches **118**. Accordingly, the AP **104** designates L3 cache **118(2)** as belonging to a first level in the L3 cache hierarchy and assigns a first FIFO pointer separation to L3 cache **118(2)**. The AP **104** designates L3 caches **118(1)** and L3 cache **118(3)** to a second level in the L3 cache hierarchy since they are both an equal distance from the shared PHY interface point **130** but farther than L3 cache **118(2)**. The AP **104** also assigns a second FIFO pointer separation to the second level of the L3

cache hierarchy, i.e., to L3 caches **118(1)** and **118(3)**, that is longer than the first FIFO pointer separation. Thus, L3 cache **118(2)** has a shorter FIFO pointer duration than L3 caches **118(1)** and **118(3)**.

(19) In this manner, the AP **104** provides a physically aware pointer separation for different ultra-short reach (USR) PHY instances between the processor core die **120** and the L3 caches **118**, and, therefore, different roundtrip latencies to the L3 caches (or other LLCs) based on the clock structure of the AP **104**. The AP **104** uses these variable latencies to implement the NUCA techniques at the L3 cache as described herein to reduce the average L3 cache access latency time. For example, the AP **104** leverages the different levels in the L3 cache hierarchy by allocating frequently used or critical applications/data to the first level of the L3 cache hierarchy (i.e., to L3 cache **118(2)** in the above example) and less frequently used or non-critical applications/data to the second level of the L3 cache hierarchy (i.e., to L3 caches **118(1)** and **118(3)** in the above example). Accordingly, the latency times associated with more frequently used or critical applications is reduced, thereby improving overall processing system **100** performance.

(20) FIG. 2 shows an example of an AP **204** implementing an NUCA L3 cache hierarchy in accordance with various implementations. Put differently, the NUCA L3 cache hierarchy is a sub-hierarchy expressly for the L3 cache compared to the cache hierarchy that is applied to the L1, L2, and L3 caches. In some implementations, the AP **204** illustrated in FIG. 2 corresponds to the AP **104** of FIG. 1.

(21) As illustrated in FIG. 2, the processor core die (PCD) **220** includes a plurality of CUs **212(1)** to **212(N)**. In some implementations, each of the CUs **212(1)** to **212(N)** corresponds to the CUS **112(1)** to **112(N)** of FIG. 1. Accordingly, each of the CUs **212(1)** to **212(N)** includes a designated L1 cache (not shown). As further illustrated in FIG. 2, the PCD **220** also includes a plurality of L2 caches **216(1)** to **216(K)**. In some implementations, the plurality of L2 caches **216(1)** to **216(K)** corresponds to L2 caches **116(1)** to **116(K)** of FIG. 1.

(22) In addition to including the PCD **220**, the AP **204** also includes a plurality of memory cache dies (MCDs) **218(1)** to **218(6)**. In some AP **204** configurations, the total number of MCDs is less than or greater than the six MCDs shown in FIG. 2. In some implementations, the L3 cache of the AP **204** is implemented at the collection of MCDs **216-1** to **216-6**. That is, each of the MCDs **216(1)** to **216(6)** holds an L3 cache. Furthermore, each of the MCDs **216(1)** to **216(6)** includes a die-to-die interface through a shared PHY interface point with the PCD **220**. In the illustration shown in FIG. 2, the shared PHY interface point (IP) is depicted as PHY IP **230**. Although shown as being separate from PCD **220**, in some implementations, the PHY IP **230** is integrated into PCD **220**. In some implementations, the PHY IP **230** is a SERDES circuitry with a clock domain crossing structure. The clock domain crossing structure may include, in some configurations, FIFO queues for processing the L3 cache access requests to the MCDs **218** based on a system on chip (SOC) clock (not shown) on the AP **204**. One example of a FIFO queue **232** implemented at the PHY IP **230** is shown in FIG. 2. In some implementations, the FIFO queue **232** is a data structure for organizing the execution of write and read requests to the L3 caches in the different MCDs **218** and includes a write pointer **234** and a read pointer **236**. The write pointer **234** is incremented when an item is written to the FIFO queue **232**. That is, the write pointer **234** references the next position in the FIFO queue **232** to be filled (i.e., written). Similarly, when an item is read from the FIFO queue **232**, the read pointer **236** is incremented. That is, the read pointer **236** references the next position in the FIFO queue **232** to be read from. The distance between the two pointers **234**, **236** is referred to as the FIFO pointer separation **238** (also referred to as FIFO write-read pointer separation, FIFO read separation, pointer separation, or the like).

(23) Conventional caching techniques treat the collection of L3 caches, and therefore each of the MCDs **218(1)** to **218(6)**, equally based on a common FIFO pointer separation. That is, conventional caching techniques do not factor into account the different physical placements of the MCDs **218** on the AP **204** and assume the same FIFO pointer separation for MCD **216(1)** as

assumed for MCD (5), for example. This leads to higher write-to-read pointer separation, or higher latencies for clock-domain crossings, at the L3 cache.

(24) On the other hand, the techniques described herein recognize the differences in the positions of the MCDs **218** on the AP **204** and exploit this information to differentiate the MCDs **218** to create an NUCA L3 cache hierarchy (also referred to as an LLC hierarchy). In some implementations, an L3 cache controller (L3 CC) **280** employs this information of the physical placement of the MCDs **218(1)** to **218(6)** to set the FIFO pointer separation in the FIFO queues in the PHY IP **230** for the corresponding MCDs **218**. This FIFO pointer separation directly impacts the die-to-die interface (e.g., a USR interface between the PCD **220** and a corresponding one of the MCDs **218** of AP **204**) roundtrip latency. In some implementations, the L3 CC **280** sets the FIFO pointer separation at the PHY IP **230** for every different position of the MCDs. That is, a different FIFO pointer separation is set for every distinct USR-PHY distance with the MCDs **218**.

(25) To illustrate this concept with respect to FIG. 2, MCD **218(1)** and MCD **218(2)** are at a first interface distance or length **250** (only one instance labeled for clarity) from the PHY IP **230** of the PCD **220**. MCD **218(3)** and MCD **218(4)** are at a second interface distance or length **260** that is longer than the first interface distance **250**. MCD **218(5)** and MCD **218(6)** are at a third interface distance or length **270** that is longer than the second interface distance **260**. Thus, a first FIFO pointer separation is designated for MCDs **218(1)** and **218(2)**, a second FIFO pointer separation is designated for MCDs **218(3)** and **218(4)**, and a third FIFO pointer separation is designated for MCDs **218(5)** and **218(6)**, where the first FIFO pointer separation is less than the second FIFO pointer separation, which in turn, is less than the third FIFO pointer separation. This in turn impacts the round trip latency for cache accesses to the respective MCDs **218**. Table I is an example table summarizing the latencies (in nanoseconds (ns)) for the different MCDs **218** for AP **204**.

(26) TABLE-US-00001

MCDs	MCDs 218(1)&(2)	MCDs 218(3)&(4)	MCDs 218(5)&(6)
FIFO pointer separation	0.2 ns	0.4 ns	0.6 ns
Impact on round trip latency	0.8 ns	1.6 ns	2.4 ns

(27) Thus, as shown in Table I, by setting a different FIFO pointer separation based on the physical location of the MCDs **218** with respect to the PCD **220**, the AP **204**, in effect, realizes different memory access times to the different ones of the MCDs **218**. The L3 CC **280** leverages these different memory access times to create an NUCA L3 cache hierarchy in which the MCDs with the fastest round trip latencies are assigned to higher levels in the hierarchy.

(28) FIG. 3 shows an example of such a hierarchy based on the discussion above with respect to FIG. 2. As shown in FIG. 3, MCDs **218(1)** and **218(2)** are assigned to the first level (or highest level) **302** of the L3 cache hierarchy, MCDs **218(3)** and **218(4)** are assigned to the second level (or middle level) **304** of the L3 cache hierarchy, and MCDs **218(5)** and **218(6)** are assigned to the third level (or lowest level) **306** of the L3 cache hierarchy. The collection of levels **302** to **306** is collectively referred to as L3 cache levels, LLC levels, an LLC hierarchy, or the like. That is, the MCDs **218** are ranked according to their corresponding roundtrip latency times with the PCD **220**. Thus, the L3 CC **280** leverages this non-uniform cache access (NUCA) architecture (non-uniform in the sense that the MCDs have different memory access times associated with them) to reduce the overall average L3 latency by placing more critical or more frequently used data blocks at the first level **302** L3 caches, and the least critical or least frequently used data blocks at the third level **306** of L3 caches. In some implementations, this NUCA L3 cache hierarchy is provided to higher levels of software or other applications running on the AP **204** or at the CPU (such as CPU **102**) so that they can allocate data to the MCDs **218** based on NUCA L3 cache hierarchy aware data allocation policies. For example, for data or application allocated to an L3 cache of the AP **204**, applications or data that are more critical or more frequently accessed by the CUs **212** are allocated to the first level MCDs (i.e., MCDs **218(1)** and **218(2)**). This reduces the latencies associated with L3 cache accesses, thereby improving overall system performance.

(29) FIG. 4 shows an example of how the techniques described herein can be used to determine which MCD positions to populate for lower L3 cache processing devices in die harvesting, in

accordance with various implementations. In some implementations, the accelerated processor (AP) **404** shown in FIG. **4** is similar to AP **104**, **204**, **304** but AP **404** is a lower product configuration with a reduced L3 cache size.

(30) As illustrated in FIG. **4**, the AP **404** includes a processor core die (PCD) **420** including a number of compute units (CUs) **412(1)** to **412(N)** and a number of L2 caches **416(1)** to **416(K)**. In some implementations, the CUs **412** each includes a private L1 cache (not pictured). The PCD **420**, in some implementations, also includes an L3 cache controller (CC) **480**. A PHY IP **430** couples the PCD **420** with each of the MCDs **418**.

(31) In this example, this reduced cache size results in four MCD positions needing to be populated. In other implementations, the reduced cache size is a different number, e.g., two MCD positions are populated. By using the techniques described herein, the MCDs at position **418-5** and **418-6** are not populated (i.e., left empty, indicated by x-signs **450**) since their associated cache latencies are higher than the other four MCDs **418-1** to **418-4**. That is, the different MCD positions **418** are grouped into levels of an LLC hierarchy based on their corresponding physical attributes. In some implementations, the corresponding physical attribute is one or more of a die-to-die interface length between a memory cache die position **418** and the processor core die **420**, or a physical location of the memory cache die position **418** on the AP **404**. A first level of the LLC hierarchy includes a subset of the plurality of memory cache die positions having a shortest die-to-die interface length between the memory cache die position **418** and the processor core die **420**, or a respective physical location closest to the processor core die **420** on the AP **404**. For example, in FIG. **4**, the positions associated with MCD **418(1)** and MCD **418(2)** are populated with MCDs prior to the positions associated with other ones of the MCDs **418**. This results in improved L3 cache latencies for lower product configuration implementations compared to the conventional manner of die harvesting which does not differentiate between the MCDs implementing the L3 cache. Therefore, the techniques described herein can be used in die harvesting (i.e., selecting which die or chiplet positions to populate) to improve the overall performance of accelerated processors with reduced L3 cache sizes.

(32) Any of the various elements or components shown in FIGS. **1-4**, including the MC **110**; the L3 CC **280**, **480**; the PHY IP **230**, **430**; and any other components associated with the APs **104**, **204**, **404** illustrated in FIGS. **1-4** are, in various implementations, implemented as one of software executing on a processor, hardware that is hard-wired (e.g., circuitry) to perform the various operations described herein, or a combination thereof.

(33) FIG. **5** illustrates an example flowchart **500** describing a method for implementing NUCA L3 cache techniques in accordance with various implementations. For example, the NUCA L3 cache techniques described herein are applied by an accelerated processor (AP) having its L3 cache on different dies or chiplets than a processor core die including the processor cores. For example, the AP includes a processor core die with the processor cores (also referred to as CUs) as well as the L1 and L2 caches and a plurality of memory cache dies (MCDs) separate from the processor core die, where the MCDs include the L3 cache.

(34) At block **502**, the AP identifies the different MCDs and their corresponding die-to-die interface lengths with the processor core die. For example, with reference to FIG. **2**, the AP **204** identifies a first die-to-die interface length **250** for MCDs **218(1)** and **218(2)**, a second die-to-die interface length **260** for MCDs **218(3)** and **218(4)**, and a third die-to-die interface length **270** for MCDs **218(5)** and **218(6)**.

(35) At block **504**, the AP assigns different FIFO pointer separations for the FIFO queues issuing the cache accesses to the L3 cache based on the die-to-die interface lengths identified at block **502**. For example, with reference to FIG. **2** and Table I, the AP **204** assigns FIFO pointer separations for the FIFO queues associated with each of the MCDs **218** based on the values shown in Table I. That is, the AP **204** assigns a FIFO pointer duration of 0.2 ns to MCDs **218(1)** and **218(2)**, a FIFO pointer duration of 0.4 ns to MCDs **218(3)** and **218(4)**, and a FIFO pointer duration of 0.6 ns to

218(5) and **218(6)**. In some implementations, by assigning different FIFO pointer durations to the MCDs, the impact on the round trip latencies of the cache access requests to the respective MCDs is tiered.

(36) At block **506**, the AP creates an L3 cache hierarchy based on the assigned FIFO pointer durations. That is, the AP groups or divides the MCDs based on the different tiers of round trip latencies created at block **504**. For example, with reference to FIGS. **2** and **3**, MCDs **218(1)** and **218(2)** have the shortest round trip latency time, so the AP **204** assigns them to a first level (e.g., 1st level **302**) of an L3 cache hierarchy. MCDs **218(3)** and **218(4)** have the next shortest round trip latency time, so the AP **204** assigns them to the second level (e.g., 2nd level **304**) of the L3 cache hierarchy. MCDs **218(5)** and **218(6)** have the longest round trip latency time, so the AP **204** assigns them to the third level (e.g., third level **306**) of the L3 cache hierarchy. The first level of the cache hierarchy, for example, is reserved for critical or more frequent access data since the round trip latency time to access the first level of the L3 cache hierarchy is less than the other levels of the L3 cache hierarchy.

(37) After the cache hierarchy is created, at block **508**, the AP issues an L3 cache access request. For example, with reference to FIGS. **1** and **2**, AP **104/204** issues an L3 cache access request based on an application running on a CPU (such as CPU **102** in FIG. **1**) for which AP **104/204** is executing associated operations.

(38) At block **510**, the AP identifies a latency associated with the L3 cache access request of block **508**. For example, with reference to FIGS. **1** and **2**, AP **104/204** identifies that the L3 cache request is associated with a critical or frequently used application or data block. Accordingly, the AP determines that the L3 caches access requires the lowest latency.

(39) At block **512**, the AP directs the L3 cache access to the L3 cache MCD based on the L3 cache hierarchy. For example, with reference to FIGS. **2** and **3**, the AP **204** directs the L3 cache access to the first level **302** in the L3 cache hierarchy including MCDs **218(1)** and **218(2)** if the AP **204** identifies that the L3 cache access request is identified as requiring the lowest latency at block **510**. Thus, by directing the L3 cache access (e.g., write or read) to one of or both of MCDs **218(1)** and **218(2)**, the AP **204** reduces the round trip latency time for executing the L3 cache access. This reduction in L3 cache latency improves system performance. The method further includes repeating blocks **508-512** for other L3 cache access requests.

(40) In some implementations, the apparatus and techniques described above are implemented in a system including one or more integrated circuit (IC) devices (also referred to as integrated circuit packages or microchips), such as the AP or its components described above with reference to FIGS. **1-4**. Electronic design automation (EDA) and computer aided design (CAD) software tools may be used in the design and fabrication of these IC devices. These design tools typically are represented as one or more software programs. The one or more software programs include code executable by a computer system to manipulate the computer system to operate on code representative of circuitry of one or more IC devices so as to perform at least a portion of a process to design or adapt a manufacturing system to fabricate the circuitry. This code can include instructions, data, or a combination of instructions and data. The software instructions representing a design tool or fabrication tool typically are stored in a computer readable storage medium accessible to the computing system. Likewise, the code representative of one or more phases of the design or fabrication of an IC device may be stored in and accessed from the same computer readable storage medium or a different computer readable storage medium.

(41) A computer readable storage medium may include any non-transitory storage medium, or combination of non-transitory storage media, accessible by a computer system during use to provide instructions and/or data to the computer system. Such storage media can include, but is not limited to, optical media (e.g., compact disc (CD), digital versatile disc (DVD), Blu-Ray disc), magnetic media (e.g., floppy disc, magnetic tape, or magnetic hard drive), volatile memory (e.g., random access memory (RAM) or cache), non-volatile memory (e.g., read-only memory (ROM) or

Flash memory), or microelectromechanical systems (MEMS)-based storage media. The computer readable storage medium may be embedded in the computing system (e.g., system RAM or ROM), fixedly attached to the computing system (e.g., a magnetic hard drive), removably attached to the computing system (e.g., an optical disc or Universal Serial Bus (USB)-based Flash memory) or coupled to the computer system via a wired or wireless network (e.g., network accessible storage (NAS)).

(42) In some implementations, certain aspects of the techniques described above may be implemented by one or more processors of a processing system executing software. The software includes one or more sets of executable instructions stored or otherwise tangibly embodied on a non-transitory computer readable storage medium. The software can include the instructions and certain data that, when executed by the one or more processors, manipulate the one or more processors to perform one or more aspects of the techniques described above. The non-transitory computer readable storage medium can include, for example, a magnetic or optical disk storage device, solid state storage devices such as Flash memory, a cache, random access memory (RAM) or other non-volatile memory device or devices, and the like. The executable instructions stored on the non-transitory computer readable storage medium may be in source code, assembly language code, object code, or other instruction format that is interpreted or otherwise executable by one or more processors.

(43) Note that not all of the activities or elements described above in the general description are required, that a portion of a specific activity or device may not be required, and that one or more further activities may be performed, or elements included, in addition to those described. Still further, the order in which activities are listed is not necessarily the order in which they are performed. Also, the concepts have been described with reference to specific implementations. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the present disclosure as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of the present disclosure.

(44) Benefits, other advantages, and solutions to problems have been described above with regard to specific implementations. However, the benefits, advantages, solutions to problems, and any feature(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential feature of any or all the claims. Moreover, the particular implementations disclosed above are illustrative only, as the disclosed subject matter may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. No limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore evident that the particular implementations disclosed above may be altered or modified and all such variations are considered within the scope of the disclosed subject matter. Accordingly, the protection sought herein is as set forth in the claims below.

Claims

1. A method comprising: issuing a cache access request to a last level cache (LLC) in a cache hierarchy of an accelerated processor, the LLC located on a plurality of memory cache dies separate from a processor core die of the accelerated processor; and based on a latency of the cache access request, directing the cache access request to a subset of the plurality of memory cache dies.
2. The method of claim 1, wherein directing the cache access request to the subset of the plurality of memory cache dies is based on the subset's corresponding die-to-die interface length with the processor core die.
3. The method of claim 2, further comprising: assigning pointer separations to each of the plurality

of memory cache dies, wherein the pointer separations are associated with queues for issuing access requests to each of the plurality of memory cache dies.

4. The method of claim 3, wherein the plurality of memory cache dies is configured as an LLC hierarchy, wherein each LLC level of a plurality of LLC levels of the LLC hierarchy corresponds to a different subset of the plurality of memory cache dies.

5. The method of claim 4, wherein each subset of the plurality of memory cache dies is assigned to one of the plurality of LLC levels of the LLC hierarchy based on the assigned pointer separations, wherein a first level of the LLC hierarchy is assigned to a subset of the plurality of memory cache dies with a lowest pointer separation.

6. The method of claim 2, wherein the corresponding die-to-die interface length is based on determining an interface length between each memory cache die of the plurality of memory cache dies and a shared physical interface point in the accelerated processor.

7. The method of claim 1, wherein the processor core die comprises a plurality of compute units and one or more higher cache levels of the cache hierarchy, wherein the higher cache levels comprise one or more level 1 (L1) caches and one or more level 2 (L2) caches, and wherein the LLC is a level 3 (L3) cache.

8. The method of claim 7, wherein the plurality of memory cache dies is arranged adjacent to the processor core die on the accelerated processor.

9. The method of claim 8, wherein the plurality of memory cache dies is divided into a plurality of subsets of memory cache dies based on their physical location on the accelerated processor, the plurality of subsets of memory cache dies comprising the subset of the plurality of memory cache dies.

10. An accelerated processor comprising: a processor core die; a plurality of memory cache dies coupled to the processor core die, the plurality of memory cache dies comprising a last level cache (LLC); and an LLC controller to: issue a cache access request to the LLC; and based on a latency of the cache access request, direct the cache access request to a subset of the plurality of memory cache dies.

11. The accelerated processor of claim 10, further comprising: a shared physical interface point between the plurality of memory cache dies and the processor core die.

12. The accelerated processor of claim 11, the shared physical interface point comprising serializer/deserializer circuitry comprising queues for issuing cache access requests to the LLC on the plurality of memory cache dies.

13. The accelerated processor of claim 12, wherein the queues are first in, first out (FIFO) queues.

14. The accelerated processor of claim 12, the LLC controller to: based on each of the plurality of memory cache dies and their corresponding die-to-die interface length with the processor core die, assign pointer separations to each of the plurality of memory cache dies, wherein the pointer separations are associated with queues for issuing access requests to the LLC on the plurality of memory cache dies.

15. The accelerated processor of claim 14, the LLC controller to: direct the cache access request to the subset of the plurality of memory cache dies based on an LLC hierarchy comprising a plurality of LLC levels, each LLC level of the plurality of LLC levels comprising a different subset of the plurality of memory cache dies, wherein each subset of the plurality of memory cache dies is assigned to one of the plurality of LLC levels of the LLC hierarchy based on the assigned pointer separations, wherein a first level of the LLC hierarchy is assigned to a subset of the plurality of memory cache dies with a lowest assigned pointer separation.

16. A non-transitory computer readable medium embodying a set of executable instructions, the set of executable instructions to manipulate at least one processor to: issue a cache access request to a last level cache (LLC) in a cache hierarchy of an accelerated processor, the LLC located on a plurality of memory cache dies separate from a processor core die of the accelerated processor; and based on a latency of the cache access request, direct the cache access request to a subset of the

plurality of memory cache dies.

17. The non-transitory computer readable medium of claim 16, wherein directing the cache access request to the subset of the plurality of memory cache dies is based on the subset's corresponding die-to-die interface length with the processor core die.

18. The non-transitory computer readable medium of claim 17, further including instructions to manipulate the at least one processor to: assign pointer separations to each of the plurality of memory cache dies, wherein the pointer separations are associated with queues for issuing access requests to each of the plurality of memory cache dies.

19. The non-transitory computer readable medium of claim 18, wherein the plurality of memory cache dies is configured as an LLC hierarchy, wherein each LLC level of a plurality of LLC levels of the LLC hierarchy corresponds to a different subset of the plurality of memory cache dies.

20. The non-transitory computer readable medium of claim 19, wherein each subset of the plurality of memory cache dies is assigned to one of the plurality of LLC levels of the LLC hierarchy based on the assigned pointer separations, wherein a first level of the LLC hierarchy is assigned to a subset of the plurality of memory cache dies with a lowest pointer separation.
